

Summary_Report1

1 总体说明

完成了 C_lab 和 Python_lab 中的所有考核。

2 运行结果说明与展示

2.1 C_lab

2.1.1 ex 1_hello

```
chen@chen-Legion-Y7000P-IRX10:~/code/FOCUS/阶段一/task2_lab/C_lab$ ls
ex1          ex2_pointers_and_functions.c  ex4          ex5_strings.c
ex1_hello.c  ex3                          ex4_pointer_arithmetic.c  ex6
ex2          ex3_arrays.c                 ex5          ex6_structs.c
chen@chen-Legion-Y7000P-IRX10:~/code/FOCUS/阶段一/task2_lab/C_lab$ ./ex1
Hello World
chen@chen-Legion-Y7000P-IRX10:~/code/FOCUS/阶段一/task2_lab/C_lab$
```

2.1.2 ex 2_pointers_and_functions

```
chen@chen-Legion-Y7000P-IRX10:~/code/FOCUS/阶段一/task2_lab/C_lab$ ./ex2
add_one: 5
add_one_pointer: 6
add_one_double_ptr: 7
chen@chen-Legion-Y7000P-IRX10:~/code/FOCUS/阶段一/task2_lab/C_lab$
```

2.1.3 ex 3_arrays

```
chen@chen-Legion-Y7000P-IRX10:~/code/FOCUS/阶段一/task2_lab/C_lab$ ./ex3
address of the start of the array: 0x7fffe2d342b4
address of index 2: 0x7fffe2d342b6
value at index 2: 10
chen@chen-Legion-Y7000P-IRX10:~/code/FOCUS/阶段一/task2_lab/C_lab$
```

2.1.4 ex 4_poniters_and_functions

```
chen@chen-Legion-Y7000P-IRX10:~/code/FOCUS/阶段一/task2_lab/C_lab$ ./ex4
address of the start of the array: 0x7ffc00108f20
address of index 2: 0x7ffc00108f24
value at index 2: 10
chen@chen-Legion-Y7000P-IRX10:~/code/FOCUS/阶段一/task2_lab/C_lab$ ./ex5
```

ex 5_strings

```
chen@chen-Legion-Y7000P-IRX10:~/code/FOCUS/阶段一/task2_lab/C_lab$ ./ex5
prints hello: hello
length of hello: 5
prints world: world
address of world_str: 0x7ffeb72c9ea6
address of 'r': 0x7ffeb72c9ea8
prints hello world: hello world
```

2.1.5 ex 6_structs

```
chen@chen-Legion-Y7000P-IRX10:~/code/FOCUS/阶段一/task2_lab/C_lab$ ./ex6
Size of a struct Student: 16 bytes
Student's ID: 5
```

2.2 Python_lab

2.2.1 TODO 1

```
# TODO: slice the first three elements of list 'y' and
# store in a new list, then print the 2nd element of this
# new list
z = y[:3]
print(z[1])
```

0

2.2.2 TODO 2

```
robots = ['baxter', 'sawyer', 'turtlebot']    "turtlebot": Unknown word.

# TODO: Using a for loop and the python built-in enumerate function,
# Print "#1: baxter", "#2: sawyer", "#3: turtlebot",    "turtlebot": Unknown word.
# each on its own line
for i,robot in enumerate(robots):
    print(f'#{i+1}: {robot}')
```

Pyt

```
#1: baxter
#2: sawyer
#3: turtlebot
```

2.2.3 TODO 3

```
A = np.array([[5, 6, 8], [2, 4, 5], [3, 1, 10]])
B = np.array([[3, 5, 0], [3, 1, 1]])
# TODO: multiply matrix A with matrix B padded with 1's to the
# same dimensions as A; sum this result with the identity matrix "identity": Misspelled
# (you may find np.concatenate, np.vstack, np.hstack, or np.eye useful).
# Make sure you don't alter the original contents of B. Print the result
row_ones = np.ones((1,B.shape[1]))
B_padded = np.vstack([B,row_ones])
C = A @ B_padded
print(C)
```

Python

```
[[41. 39. 14.]
 [23. 19.  9.]
 [22. 26. 11.]]
```

2.2.4 TODO 4

```
# TODO: print every other integer from 0 ... 6 multiplied by 2
# as a list
result = [x*2 for x in range(0,7,2)]
result
```

Pyth

```
[0, 4, 8, 12]
```

2.2.5 hub 部分

完成了代码部分的编程，借助 AI 完成逻辑测试与验证。

注：学习记录与日志记录在

阶段一/学习日志/学习日志.txt 下